

Executive Scenario Report

Predicted Load	Uncertainty	Financial Exposure	Grid Status
152.6 MWh	+/- 3.7 MWh	\$20,449	CRITICAL

Strategic Overview

This is a short-term operational energy forecast predicting a value of approximately 152.6 units with a low uncertainty of 3.7. The ensemble model heavily relies on CatBoost, and the prediction is primarily driven by time-of-day effects and recent historical demand patterns. This forecast is crucial for immediate operational planning and resource dispatch.

Critical Risk Factors

- Unforeseen rapid weather changes not captured by the 'weather_temp' feature (which shows 0 contribution) could significantly impact demand.
- Sudden grid events or equipment failures could cause actual demand to deviate sharply from the prediction, leading to operational imbalances.
- The high reliance on the CatBoost model (65% weight) introduces a single point of failure risk if that model's underlying assumptions or data inputs become compromised.

Actionable Recommendations

- Continuously monitor real-time weather conditions and integrate more granular, real-time weather data into the forecasting model to improve accuracy.
- Implement dynamic contingency reserves and flexible generation assets to quickly respond to unexpected demand fluctuations or supply disruptions.
- Regularly re-evaluate and re-optimize the ensemble model weights, especially for the highly weighted CatBoost model, to ensure robustness and adapt to evolving energy consumption patterns.

Input Feature Context

Hour:	8
Day Of Week:	2
Month:	1
Is Weekend:	0
Lag 24H:	134.4
Lag 12H:	130.3

Roll Mean 24H:	137.8
Roll Std 24H:	5